

# **AI-Powered Personalized Course Creation and Discovery Platform with Adaptive Learning Paths**

R26-SE-041

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in  
Information Technology

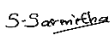
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March 2026

## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

| Name       | Student ID | Signature                                                                           |
|------------|------------|-------------------------------------------------------------------------------------|
| Sarmitha S | IT22637482 |  |

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

| Name                 | Date       | Signature                                                                             |
|----------------------|------------|---------------------------------------------------------------------------------------|
| Prof. Nuwan Kodagoda | 15/03/2026 |  |

## ABSTRACT

In the current educational environment, students often rely on handwritten notes for learning and exam preparation. However, handwritten notes vary in quality, format, and language, which makes them difficult to digitize, search, or share efficiently. Many students struggle to organize their notes in a way that enhances study effectiveness.

Existing tools for note digitization primarily focus on high-quality printed documents or single-language notes. There is limited research on creating an end-to-end system that can handle handwritten notes of varying quality, support multilingual text extraction, and automatically structure notes for easy study. Additionally, automated flashcard generation from handwritten notes is largely unexplored.

This project proposes a complete pipeline for converting handwritten notes into study-ready digital resources. The system preprocesses images, extracts text using OCR, translates content, structures the notes with headings and keywords, and generates automatic flashcards. This approach ensures that students can efficiently access organized, searchable, and interactive learning materials.

The solution integrates OCR engines for text extraction, machine translation models for multilingual support (Sinhala ↔ Tamil / Sinhala ↔ English), natural language processing for text structuring, and algorithms for automatic flashcard generation.

The system is expected to convert handwritten notes into clean and structured notes. Flashcards will provide an interactive learning tool, enabling students to study more effectively. The solution will also reduce the time spent on manual note organization and revision.

This project has the potential to improve educational accessibility, especially for students in multilingual contexts. By transforming handwritten notes into digital study resources, it supports inclusive learning and enhances academic productivity. Commercially, the system can be integrated into e-learning platforms or offered as a standalone educational tool.

**Keywords:** Handwritten Notes, OCR, Multilingual Translation, Study-Ready Notes, Automatic Flashcards

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## LIST OF ABBREVIATIONS

| Abbreviation | Full Form                         |
|--------------|-----------------------------------|
| AI           | Artificial Intelligence           |
| OCR          | Optical Character Recognition     |
| NLP          | Natural Language Processing       |
| ML           | Machine Learning                  |
| API          | Application Programming Interface |
| LKR          | Sri Lankan Rupees                 |
| CNN          | Convolutional Neural Network      |
| SVM          | Support Vector Machine            |

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# 1. INTRODUCTION

## 1.1 Background and Context

In today's academic world, many students still prefer handwriting to help them remember complex concepts. However, this creates a major hurdle because physical notes cannot be searched for keywords, they take up space, and they are easily damaged. For students working across languages like Sinhala, Tamil, and English, this is even worse, as they must spend hours manually transcribing and translating their own notes before they can even begin to study effectively.

This problem primarily affects university students who need to bridge the gap between their native language notes and English academic requirements. The manual process of digitizing a notebook is a huge cognitive burden, stealing time that should be spent on actual learning. Automating the transition from paper to digital is not just a convenience; it is a necessity for modern academic success and organized knowledge management.

While local statistics are limited, global trends show that while most students find handwriting better for understanding, nearly all prefer digital formats for long-term storage. In regions using low-resource languages like Sinhala and Tamil, standard OCR tools often fail because they aren't trained on local handwriting styles or cannot handle blurry, low-quality images. This makes current solutions unreliable for a student's real-world needs.

Existing market solutions like Google Lens or standard scanners are generally designed for clear, printed text on white paper. They often fall short when faced with a student's notebook, which may contain cursive, messy layouts, or out-of-focus photos. Furthermore, these tools usually just provide a "text dump" without any organization, leaving the student with a digital mess that is just as hard to navigate as the original paper version.

The proposed solution solves this by creating an end-to-end pipeline. It uses a preprocessing module to fix blur and noise, an OCR engine to read multilingual handwriting, and a translation module for Sinhala, Tamil, and English. Finally, it uses NLP to organize the text into structured headings and automatically generates flashcards. This transforms a simple photo of a notebook into a searchable, studyready digital library.

This research aligns with UN Sustainable Development Goal 4, which promotes inclusive and high-quality education. By removing language barriers and making digitization accessible, this tool ensures that students aren't held back by the language they speak or their access to expensive tech. It uses innovation to make learning more equitable for diverse linguistic communities.

Technically, this project sits within the Natural Language Processing (NLP) and Computer Vision research cluster. It goes beyond simple text recognition to explore Knowledge Engineering, focusing on how machines can transform raw human

handwriting into structured digital assets. The goal is to preserve the benefits of handwriting while giving students the power and flexibility of a modern digital database.

## 1.2 Literature Review

Handwritten notes remain an essential part of learning in schools and universities but converting them into usable digital formats has always been challenging. Several studies have focused on extracting text from images using OCR (Optical Character Recognition). Traditional OCR techniques use preprocessing methods such as grayscale conversion, binarization, noise removal, and edge detection to enhance text visibility for recognition [10]. These approaches have proven effective for printed documents, invoices, and forms but struggle with handwritten notes, complex layouts, or low-quality images [9]. To address these limitations, some researchers have integrated convolutional neural networks (CNNs) for handwritten character recognition. CNN-based models improve the accuracy of identifying individual characters or words from handwritten text [5,6]. However, these methods primarily focus on isolated characters or small sections of text rather than complete pages, and they do not provide structured content for study purposes.

Other studies explored multilingual text recognition and translation for languages such as Sinhala and Tamil [2,3]. While these tools support digital text translation, they are not designed to work directly on images of handwritten notes. Consequently, students still need to manually digitize their notes before translating or organizing them. Some research has combined OCR with web-based interfaces, allowing users to upload images and receive text output quickly [10]. These systems are user-friendly and convenient but often output unstructured text that requires further organization. In addition, some approaches incorporate text detection techniques, such as morphological analysis, CNN-based region detection, and text localization, to improve recognition accuracy on complex or noisy images [4,6,7]. Despite these advancements, most systems either handle text extraction, multilingual translation, or character recognition independently, without integrating them into a workflow that is tailored for students' learning needs.

Furthermore, there are tools for specific tasks, such as mathematical equation recognition, which apply preprocessing, row and column segmentation, and SVM-based classification to recognize symbols accurately [11]. These methods highlight the potential for using a combination of image preprocessing, segmentation, and machine learning models to improve accuracy. Nevertheless, even these advanced methods are limited to specific types of content, such as printed equations, and cannot be directly applied to diverse, handwritten student notes. Overall, the literature shows strong capabilities in OCR, character recognition, and translation separately but lacks a comprehensive solution that combines all these functionalities to transform handwritten notes into study-ready material with structured summaries, highlighted keywords, and learning aids like flashcards.

## 1.3 Research Gap

Existing OCR and handwritten recognition systems provide partial solutions for converting text from images into digital formats, but they remain insufficient for students' study needs. Standard OCR tools perform well on printed text but often fail with handwritten notes [9,10]. Deep learning-based handwritten recognition models, such as CNNs, can identify individual characters with high accuracy [5,6], yet they are generally limited to character-level recognition rather than full-page note extraction. Multilingual translation tools can convert text between languages like Sinhala, Tamil, and English [2,3], but they require the text to be pre-digitized and structured, which adds extra effort for students. Existing web-based OCR platforms allow users to upload images and extract text quickly [10], but they provide unorganized outputs that lack any study-focused structuring or content summarization. Some methods, like mathematical symbol recognition using SVM classifiers [11], demonstrate the benefits of combining preprocessing, segmentation, and machine learning for accuracy, yet they are narrowly focused on specific tasks and do not address general handwritten notes or multilingual content.

In comparison, my proposed system addresses these gaps by offering an end-to-end student focused pipeline that integrates several functionalities into a single platform. It starts with handwritten note extraction. Next, the OCR module extracts text accurately, including text written in multiple languages. The translation module enables seamless Sinhala to Tamil and Sinhala to English conversion, which existing systems do not provide directly from images. Beyond extraction and translation, my system also structures the text into study-ready notes, organizing content by topic, highlighting keywords, and summarizing important points. Finally, it includes an automatic flashcard generation module, producing Q&A style cards for efficient learning and revision.

The novelty of this research lies in its comprehensive integration of functions, which no current system fully provides. While prior work excels in individual tasks OCR, multilingual translation, or character recognition they fail to deliver a complete workflow tailored to students' needs. This integrated approach reduces manual effort, improves learning efficiency, and transforms handwritten notes into usable study material instantly. By combining text extraction, translation, structured note creation, and flashcard generation, this system fills a critical gap in educational technology, providing a practical and innovative solution for multilingual, handwritten content in a single platform.

| Features / Systems                  | Existing Systems | My Proposed System |
|-------------------------------------|------------------|--------------------|
| Handwritten note extraction         | Partial          | ✓                  |
| Multilingual translation            | Partial          | ✓                  |
| Study-ready note structuring        | X                | ✓                  |
| Automatic flashcard generation      | X                | ✓                  |
| End-to-end student-focused pipeline | X                | ✓                  |

Table 1: Research Gap Comparison

## 2. OBJECTIVE

### 2.1 Main Objective

To develop an end-to-end system that converts handwritten notes in Sinhala language into structured, and study-ready digital notes, including multilingual translation from Sinhala to Tamil and Sinhala to English , and automatically generates flashcards for students.

### 2.2 Specific Objectives

1. Input Handling: To develop an input module that accepts handwritten notes as images.
2. Text Extraction: To design an OCR system that accurately extracts handwritten text.
3. Multilingual Translation: To integrate a translation module that converts notes between Sinhala to Tamil and Sinhala to English, ensuring accessibility across languages.
4. Notes Structuring and Cleaning: To clean and organize extracted text into structured study notes with headings, subtopics, and highlighted keywords for easy revision.
5. Flashcard Generation: To automatically generate interactive Q&A flashcards from structured notes.

### 3. METHODOLOGY

For this project, we followed a Design Science approach, which means we focused on creating a practical system for students that solves real problems. Our system is designed to take handwritten notes, convert them into digital text, translate them into multiple languages, organize them into easy-to-read study notes, and even make flashcards automatically. We built the system step by step, testing and improving each part to make sure it works well for students. The system has several parts. First, the handwritten note extraction module uses Optical Character Recognition (OCR) to read handwritten text and turn it into digital text.

Next, for translation we translate the languages from Sinhala to Tamil, and English, making it accessible to more students. After that, the study-ready note module organizes the text into clear sections, highlights important points, and formats it so it's easy to study. Finally, the flashcard module automatically creates flashcards from the key points and definitions to help students remember important information. For the technology, we chose React.js for the frontend because it loads quickly and works well on different devices. The backend is built with Fast API, which is fast and works easily with Python libraries like OCR and translation tools.

We collected handwritten notes from students with their permission and included publicly available educational materials to make our dataset diverse. To check how well the system works, we measured OCR accuracy by comparing it to manually typed text. Translation quality was checked using automatic scores and by asking experts to verify the meaning. We also asked students to try the study notes and flashcards to see if they were easy to understand and helpful for learning. This method ensures the system is simple, practical, and useful for students, making handwritten notes easier to use, learn from, and share.

## 4. HIGH-LEVEL SYSTEM ARCHITECTURE

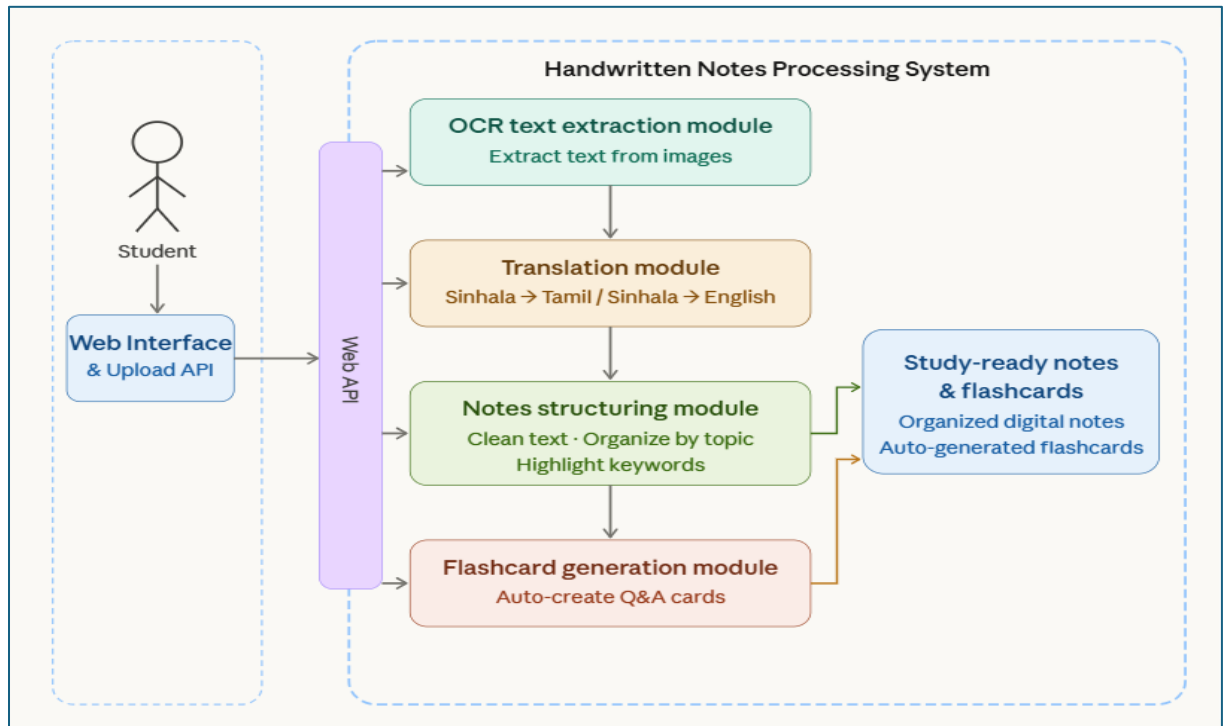


Figure 1: High Level System Architecture

## 5. USER REQUIREMENTS

### Stakeholder Identification

**Primary Users:** Students who take handwritten notes and want to convert them into digital, structured, and searchable study materials.

**Secondary Users:** Teachers and tutors who may want to access students' structured notes for teaching, feedback, or assessment purposes.

### Functional Requirements

The system shall accept handwritten note images as input.

The system shall extract text from handwritten images of Sinhala using OCR.

The system shall translate text between Sinhala, Tamil, and English as required.

The system shall clean and organize the extracted text into structured study notes with headings, subtopics, and highlighted keywords.

The system shall automatically generate flashcards with questions, answers, and key summaries.

### Non-Functional Requirements

**Performance:** The system should process a handwritten image and return structured notes and flashcards within a reasonable time.

**Usability:** The user interface should be intuitive, easy to navigate, and require minimal technical knowledge.

**Security:** Users' notes and flashcards should be securely stored with proper access control to protect privacy.

**Scalability:** The system should support multiple users and large numbers of note images without degradation in performance.

**Reliability:** The system should handle poor-quality images and continue processing without crashing.

**Compatibility:** The system should work on common devices like laptops, tablets, or smartphones that students use.

## 6. COMMERCIALIZATION PLAN

Handwritten Notes Digitization and Study Material Generation Systems that utilize Optical Character Recognition (OCR), Natural Language Processing (NLP), and Machine Learning (ML) technologies present strong commercialization potential in the education sector. The target market includes students, teachers, educational institutions, and e-learning platforms that require efficient digital study materials. The system's value proposition lies in automatically converting handwritten notes into structured, multilingual study-ready notes and flashcards, reducing manual effort and improving learning efficiency. A subscription-based and freemium revenue model can be adopted, allowing basic features for free while offering premium services such as unlimited processing, translation, and flashcard generation. The main costs involve system development, cloud storage, maintenance, and AI processing services, while pricing can be designed to remain affordable for students and scalable for institutions. The system gains competitive advantage through multilingual handwriting support, automatic note structuring, flashcard generation, and digital library integration, which are not fully provided by traditional OCR tools. Intellectual property protection can be ensured through copyrights, trademarks, and potential patents for novel processing methods. With proper marketing strategies and institutional partnerships, the system can be successfully deployed as a scalable educational technology solution.

## 7. BUDGET AND JUSTIFICATION

Developing the Handwritten Notes Digitization and Study Material Generation System is mainly a software-based implementation; therefore, no specialized hardware components are required beyond a standard personal computer used for development and testing. However, training and running OCR, Natural Language Processing (NLP), and translation models require considerable computational resources. To address this requirement, cloud computing services are planned to be used for processing handwritten images, storing generated notes, and hosting the digital library. Therefore, the primary source of cost will be the subscription or usage fees paid to cloud service providers for virtual machines, storage, and API services. Additionally, several supporting expenses such as internet usage, learning resources, and technical services are expected during the development process. The estimated budget allocation is shown below.

| Component             | Amount (LKR) |
|-----------------------|--------------|
| Internet Usage        | 3500         |
| Learning Materials    | 4000         |
| Cloud Service Charges | 4500         |
| Technical Charges     | 2000         |
| <b>Total</b>          | <b>14000</b> |

Table 2: Budget Plan

## 8. WORK BREAKDOWN STRUCTURE (WBS)

To ensure the successful delivery of the "AI-Powered Personalized Course Creation and Discovery Platform," the project is modularized into four distinct, yet interconnected, AI components. The Work Breakdown Structure (WBS) is designed to allocate an end-to-end development pipeline to each researcher, ensuring equal technical complexity and a balanced workload.

### 8.1 Task Breakdown Per Member

#### Member 1: Kojithan P.Y (IT22264220) - Interactive Infographic Generator

Task 1.1: Design a schema to extract and structure key entities, relationships, and statistics from user notes using an open-source Large Language Model (LLM). Task 1.2: Implement multiple visualization templates (hierarchical, timeline, network) using open-source libraries.

Task 1.3: Develop a dynamic interface allowing users to filter data, click elements to drill down into source quotes, and toggle visualization types.

Task 1.4: Integrate a rule-based or lightweight AI model to suggest the most suitable chart type.

Task 1.5: Enable real-time collaborative features for simultaneous viewing and annotation.

Task 1.6: Create an export module for interactive HTML or static image publishing.

#### Member 2: Baskaran V (IT22172600) – Interactive AI-Powered Audio Review Component

Task 2.1: Implement automatic speech recognition (ASR) to transcribe uploaded audio/podcasts.

Task 2.2: Develop an NLP pipeline using an open-source LLM to analyze transcripts for key concepts, structured summaries, and central themes.

Task 2.3: Build an interactive web interface with clickable transcript segments for context-specific Q&A.

Task 2.4: Integrate an open-source Text-to-Speech (TTS) model for verbal AI answers.

Task 2.5: Conduct system evaluation based on transcription accuracy, response latency, and user engagement.

### Member 3: Nishara T (IT22223876) – Adaptive Quiz and Assessment Generation Engine

Task 3.1: Develop a Knowledge Tracing Model to infer concept mastery from interaction history.

Task 3.2: Implement an NLP pipeline to auto-generate diverse question types (MCQ, fillin, short answer).

Task 3.3: Create an algorithm to dynamically adjust quiz difficulty and focus based on the learner model.

Task 3.4: Integrate a feedback mechanism offering Socratic hints and scraped web study resources.

Task 3.5: Evaluate the engine's impact on learning efficacy via pre/post-test scores.

### Member 4: Sarmitha S (IT22637482) – Multilingual Handwritten Notes Extractor

Task 4.1: Collect handwritten notes in Sinhala.

Task 4.2: Train or fine-tune an AI model for extracting the notes from images.

Task 4.3: Convert the unstructured notes into clear, and structured study based notes.

Task 4.4: Generate flashcards automatically.

### **Timeline Alignment The project spans the 2026 academic year and is divided into four synchronized phases to ensure all components integrate smoothly into the Central Knowledge Orchestrator:**

1. Phase 1: Planning & Data Collection (Months 1-2): Literature review, UI/UX design, synthetic data generation, and gathering multilingual note datasets.
2. Phase 2: Core Model Development (Months 3-5): Independent development of LLM extraction schemas, knowledge tracing models, and handwriting recognition algorithms.
3. Phase 3: Integration & UI Implementation (Months 6-7): Connecting the four standalone AI modules to the unified Personalized Learning Interface dashboard.
4. Phase 4: Testing & Pilot Study (Months 8-10): Executing the controlled pilot study with undergraduate volunteers to evaluate performance, followed by ethical clearance reporting and final documentation.

## Balanced Workload Distribution

The workload is equitably distributed by assigning each member a complete, vertically integrated subsystem. Every researcher is responsible for their module's data ingestion (e.g., text, audio, images), underlying AI/ML processing (e.g., NLP extraction, speech recognition, computer vision), user interface integration, and individual performance evaluation. This ensures that all four members engage deeply with core AI domains and software engineering practices without creating technical bottlenecks or unequal dependencies

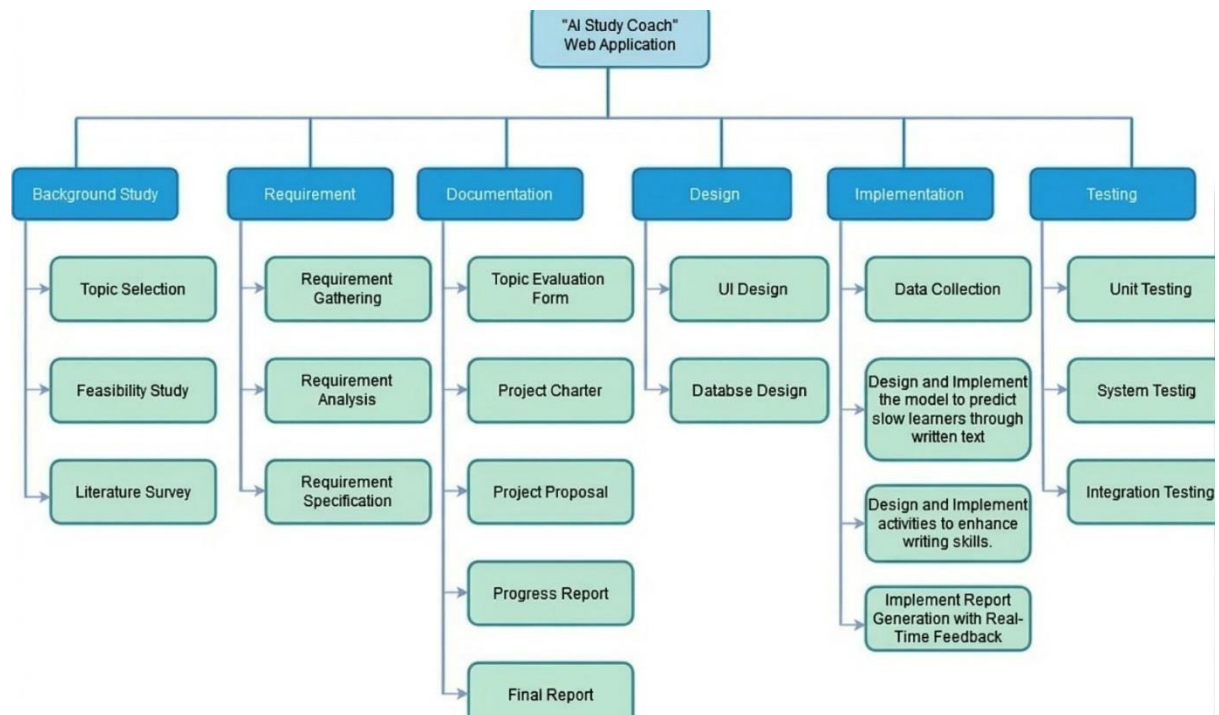


Figure 2: WBS Diagram

## 9. GANTT CHART

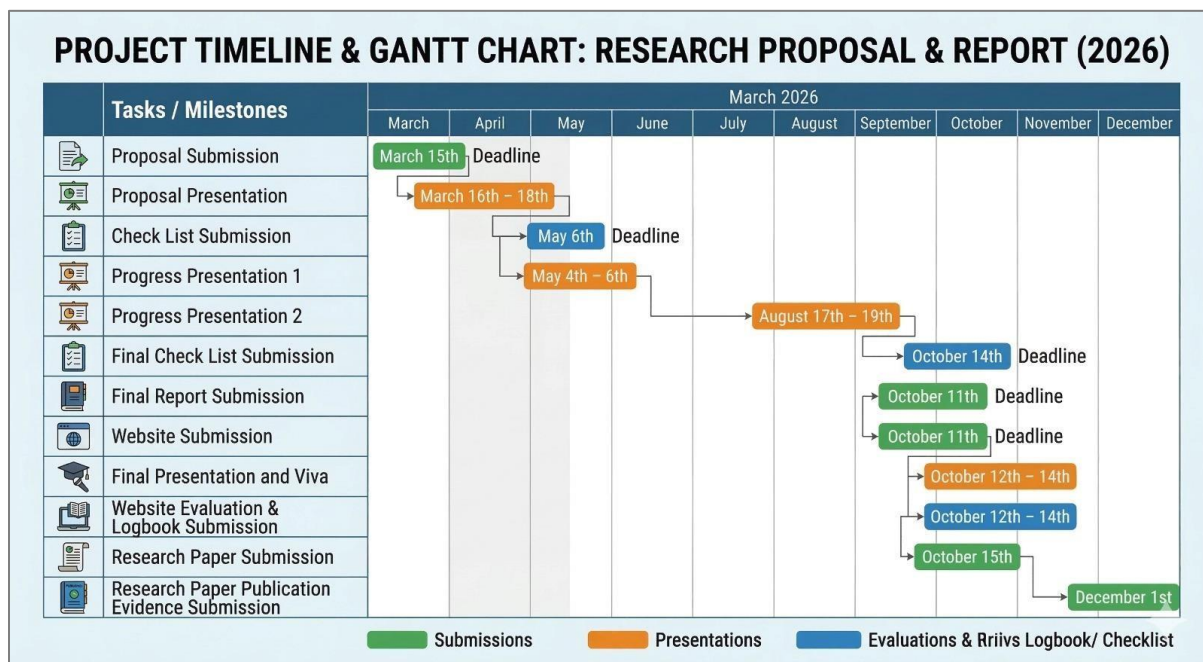


Figure 3: Gantt Chart

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## 11. APPENDICES

### Appendix A: Survey instruments

#### 1. Online Questionnaires (via Google Forms)

What it is: Digital surveys given to students before and after using your interactive charts.

Why use it: To collect hard numbers and quantitative data (e.g., rating usability on a scale of 1 to 5).

Example: "Rate how easy it was to switch from the Timeline View to the Network View."

#### 2. Structured Interviews

What it is: A one-on-one conversation using a strict, pre-written list of specific questions.

Why use it: To make sure you get qualitative answers on specific features you built (like the AI chart recommendation).

Example: "Did the AI correctly suggest the best chart type for your notes? Please explain why or why not."

#### 3. Unstructured Interviews

What it is: An open, free-flowing conversation with the user, without a strict script.

Why use it: To discover unexpected insights. Sometimes users interact with a visual chart in ways you didn't plan for. This allows the user to freely share their feelings, frustrations, or "aha!" moments.

Example: "Tell me about your experience exploring the data today. What stood out to you?" (And then you just listen and let the conversation flow naturally).

## Appendix B : Detailed diagrams

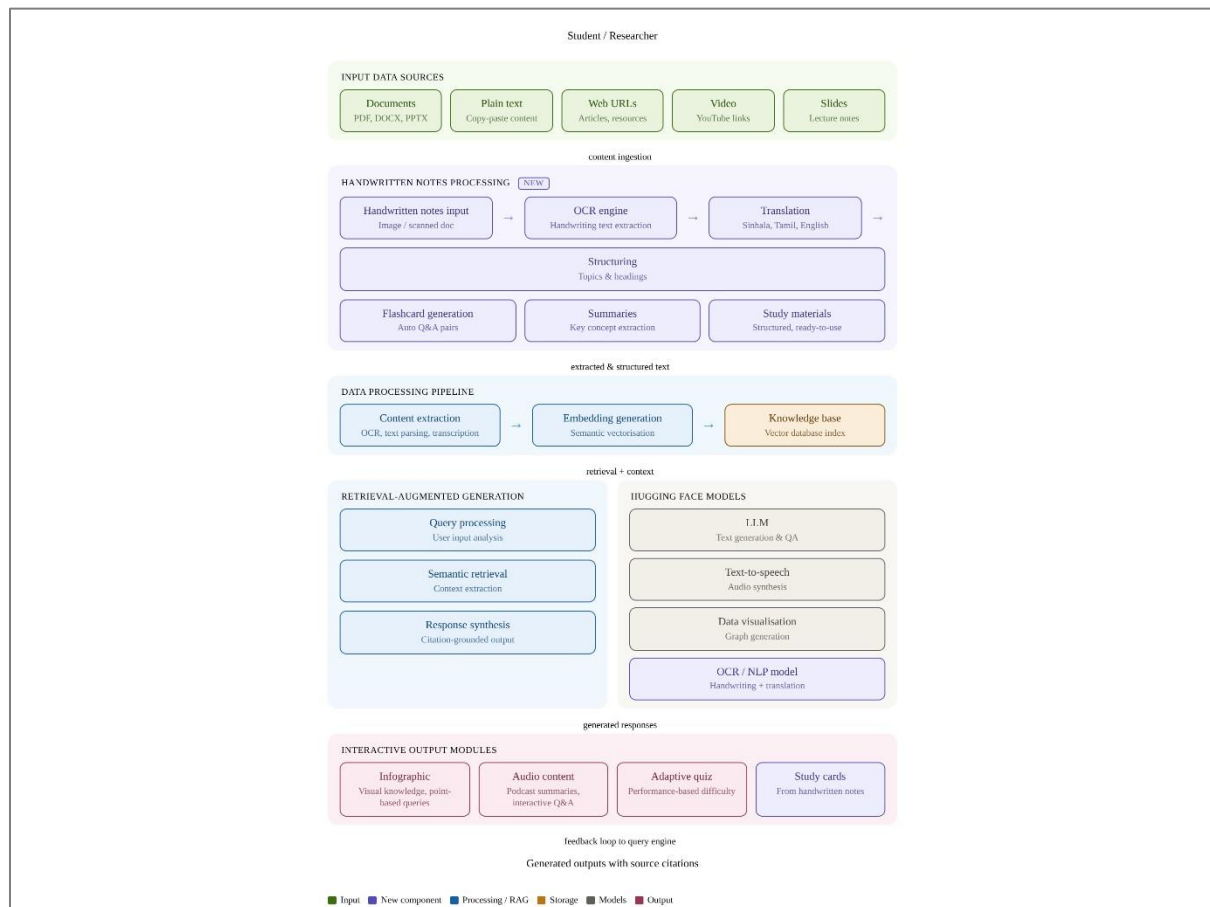


Figure 4: Detailed Diagram

## Appendix C : Preliminary UI mockups

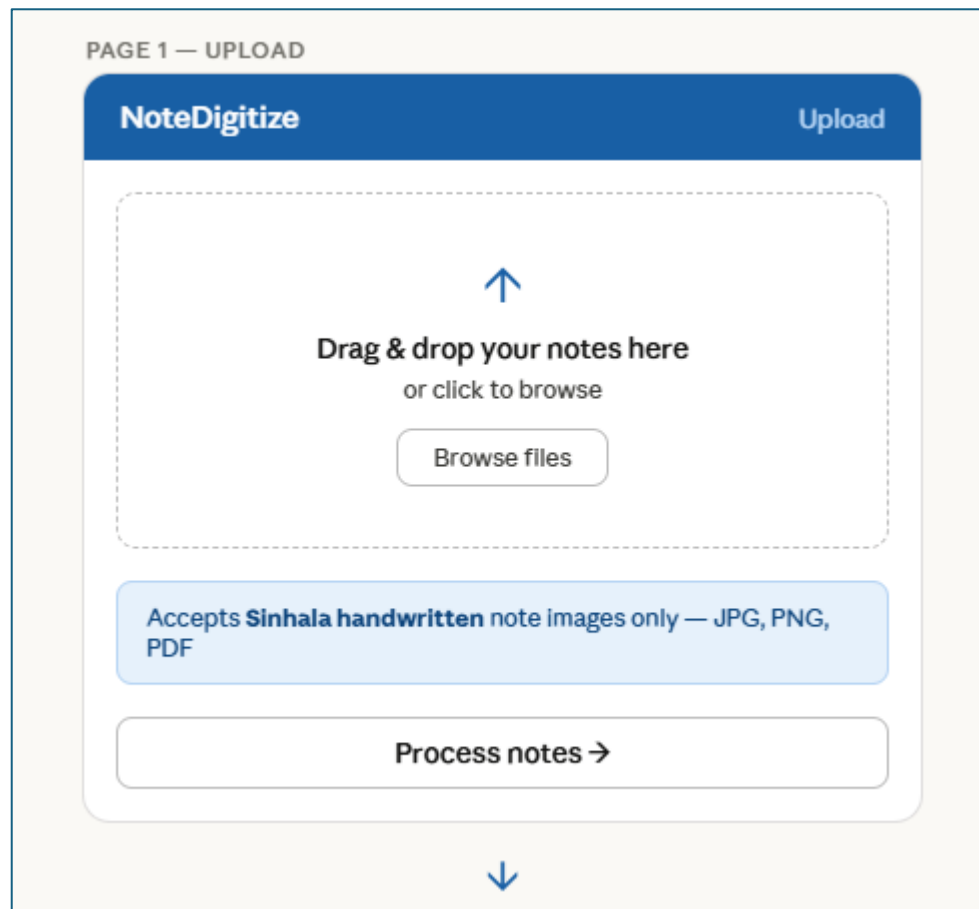


Figure 5: UI\_MockUp\_01

NoteDigitize

notes\_science.jpg

## EXTRACTED TEXT FROM YOUR IMAGE

ප්‍රකාශ සංශ්ලේෂණය යනු කොළ පැහැ ගෘහ ජූර්යාලෝකය භාවිතා කරමින් ආහාර නිපදවන ක්‍රියාවලියයි. මෙය ක්ලෝරොප්ලාස්ට් තුළ සිදු වේ. CO<sub>2</sub> සහ ජලය අවශ්‍ය වේ. ක්ලෝරොෆිල් ආලෝක ශක්තිය අවශෝෂණය කරයි.

## Select output language

Choose how you want your study notes and flashcards

Sinhala

No translation

Tamil

Si → Ta

English

Si → En

Sinhala text will be translated to **Tamil** before generating study notes and flashcards

Generate study notes &amp; flashcards →

Figure 6: UI\_MockUp\_02

## NoteDigitize

Sinhala → Tamil

Extracted text

Study notes

Flashcards

## ஒளிச்சேர்க்கை

- ஒளிச்சேர்க்கை என்பது பச்சை தாவரங்கள் சூரிய ஒளியைப் பயன்படுத்தி உணவை உற்பத்தி செய்யும் செயல்முறையாகும்
- இந்த செயல்முறை தாவர செல்களில் உள்ள குளோரோபிளாஸ்ட்களில் நடைபெறுகிறது
- கார்பன் டை ஆக்சைடு ( $\text{CO}_2$ ) மற்றும் நீர் ( $\text{H}_2\text{O}$ ) தேவைப்படுகிறது
- குளோரோபிளாஸ்டில் ஒளி ஆற்றலை உறிஞ்சுகிறது

சமன்பாடு



← Extracted

PDF

DOCX

Flashcards →

Figure 7: UI\_MockUp\_03

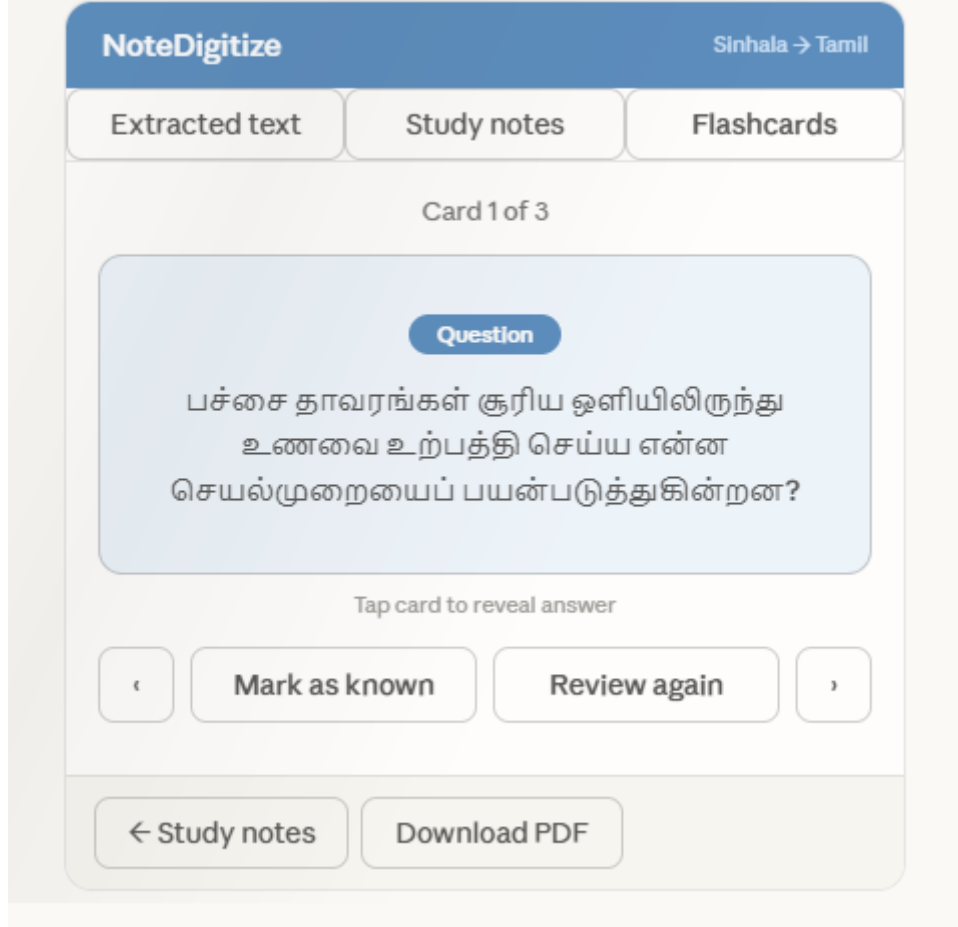


Figure 8: UI\_MockUp\_04

## Appendix D : Risk Analysis

### 1. Technical Risk: OCR Accuracy Errors

The TrOCR model may misread unclear or messy Sinhala and Tamil handwriting, extracting incorrect text or missing important words and sentences from the uploaded note image. To mitigate this risk, the TrOCR model will be fine-tuned specifically on a Sinhala and Tamil handwritten notes dataset to improve accuracy. The user interface will also allow the student to review and manually correct any extracted text before it moves to the next stage of the pipeline.

### 2. Performance Risk: Slow Processing Times

Running multiple AI models in sequence image preprocessing, OCR, translation, notes structuring, and flashcard generation may cause the system to process slowly, frustrating users who expect quick results. To mitigate this risk, all heavy AI model processing will be handled on the backend cloud server so the user's device only displays the final output. A real-time progress bar will show each processing step so the user knows the system is working and does not feel the system has frozen.

### 3. Translation Risk: Inaccurate Academic Term Translation

The mBART-50 translation model may incorrectly translate subject-specific academic terms from Sinhala or Tamil into English, causing the structured notes and flashcards to contain wrong information. To mitigate this risk, the translation module will be tested extensively using academic subject content. The user interface will allow the student to review and edit the translated text before the notes structuring stage begins, ensuring incorrect translations do not carry forward into the final output.

### 4. Quality Risk: Poor Flashcard Generation

The T5 question generation model and RoBERTa answer extraction model may produce low quality or irrelevant flashcard questions and answers if the extracted or translated text contains errors from earlier stages. To mitigate this risk, the pipeline will be carefully chained so each stage produces clean output before passing to the next. The flashcard interface will allow the student to edit, delete, or regenerate any flashcard they are not satisfied with, giving full control over the final study material.

### 5. Ethical Risk: Data Privacy Concerns

Processing student uploaded handwritten note images could accidentally collect or expose private or sensitive personal information contained within the notes. To mitigate this risk, strict data minimization will be implemented. The system will only use uploaded images for processing and will never store them permanently all uploaded files will be automatically deleted from the server immediately after processing is complete. No personal note content will be shared with any third party.